Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection

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***Abstract*—Although fingerprint biometrics are broadly used in authentication systems, they are facing a serious problem: in case the templates are stolen or compromised, the original fingerprint patterns can be restored. Such reconstruction attacks allow adversaries to create synthetic fingerprints to bypass authentication or gain access to large-scale systems. Complementary to most traditional cancelable biometric approaches aiming at template protection, we tackle for the first time the practical problem of detecting such reconstruction attempts before causing misuse. We propose a hybrid dynamic projection-based detection framework identifying inconsistencies between genuine fingerprint embeddings and synthetically reconstructed templates. Based on a ResNet-50 backbone trained with metric learning, our system extracts robust 512-D embeddings and projects them dynamically with a hybrid projection to produce protected templates. The reconstructed samples show measurable deviations in projection consistency, score distribution, and correlation structure, allowing our system to reliably detect attacks while minimally influencing normal authentication accuracy. Based on expensive experiments using the CASIA-FingerprintV5 dataset and extensive simulated attack scenarios, we explain that the proposed approach provides strong attack detection performance at improved high recognition.**

***Index Terms*—Fingerprint biometrics, template reconstruction attack, biometric security, cancelable biometrics, dynamic random projection, deep learning, ResNet-50, attack detection, biometric forensics.**

I. Introduction

Biometric authentication [1] is now a cornerstone of modern security systems due to its blend of convenience, user acceptance, and—when properly deployed—high robustness. Fingerprints occupy a particularly dominant position because of their ubiquity across consumer devices and institutional deployments. However, the permanence of biometric traits introduces a key security tension: unlike passwords or tokens, a biometric cannot easily be revoked or replaced if compromised [2]. This immutability amplifies the impact of template leakage. When an attacker obtains a stored template, it can be used as the basis for synthetic biometric generation that replicates the original identity. Recent advances in deep learning, image synthesis, and optimization-based inversion have increased the realism with which protected templates can be inverted or transformed into spoof artifacts. Template reconstruction attacks range from minutiae estimation procedures to fully learned inversion networks (including GAN- and diffusion-based approaches). These attacks do not merely expose a template—they generate usable biometric artifacts that can be replayed against sensors or used to attack other databases (cross-system transfer). Consequently, defenses that only harden templates against inversion may still fail if the attacker succeeds: the reconstructed fingerprint itself can be used in the wild. Motivated by this gap, we redirect attention from passive protection to active detection [3]: rather than only making templates hard-to-invert, we monitor and detect behavioral inconsistencies in protected templates that betray reconstruction attempts. Our key insight is that hybrid, user-specific dynamic projections introduce structural relationships between a user's embedding, its empirical correlation structure, and the randomized key component used to protect the template. Genuine biometric acquisitions preserve these relationships across multiple captures, while reconstructed templates—produced by inversion and synthesis pipelines that do not have access to the true biometric dynamics—exhibit systematic deviations. Detecting these deviations allows operators to flag and stop reconstructed-template use before compromise propagates. The contributions of this paper are threefold: 1. We introduce a hybrid dynamic projection fingerprint behavior model that fuses embeddingdriven correlation structure with user-controlled randomization to produce protected templates whose internal relationships are measurable at authentication time. 2. We design a projection-consistency detection pipeline that computes multiple consistency metrics (correlation preservation, projection variance, score distribution divergence, and embedding reconstruction residuals) and feeds the results to a lightweight classifier to decide whether an input is genuine or reconstructed. 3. We present extensive experiments on CASIA-FingerprintV5 together with simulated and learned inversion attacks, reporting detection and recognition metrics demonstrating the approach's effectiveness and operational feasibility. An important practical requirement for any detection scheme is that it should avoid substantially degrading the primary authentication task. Our experiments show that the proposed method keeps authentication performance within a small, acceptable margin while delivering strong detection results. We also discuss deployment considerations (on-device vs server-side checks), privacy implications, and adaptive attacker scenarios.

II. Related Work

Recently, there have been efforts in the area of biometrics security to enhance fingerprint authentication systems against spoofing attacks, replay attacks, and template inversion attacks. Komail Kesana et al. [4]presented a method to combine software-based spoofing detection using CNN, SVM, and ridge frequency and pore analysis to identify spoofed fingerprints created from gelatin or silicone materials. Although this method is useful in defending against spoofing attacks, it is primarily designed to provide sensor-level protection and does not counter template reconstruction attacks. Sani M. Abdullahi et al [5] presented a one-time bio- metric template system that combines deep learning, Bio- Hashing, cryptographic hashing, and symmetric encryption. This method provides an inability for an attacker to defend against replay attacks, but it does not attempt to identify reconstructed templates. Tanguy Gernot and Christophe et al [6] Rosenberger, in their paper "One-time Biometric Template-Based Approach to Prevent Replay Attacks in Traditional Systems," have suggested a one-time biometric template-based approach to prevent replay attacks in traditional systems, which uses deep learning, bio-hashing, and cryptography to provide enhanced security and accuracy in authentication.

Shah Mahmood et al [7]argued that the present condition of biometric template protection techniques is fragmented, either concentrating on standalone cryptographic methods or particular attack models without considering the regulatory and deployment viewpoints. The research work has pointed out the shortcomings in dealing with overall threats like template reversal attacks, cross-match attacks, adversarial attacks, and quantum attacks. To address the above-mentioned shortcomings, the researcher has suggested an attack-centric, cross-modal biometric template protection framework that combines technical, legal, and risk-related viewpoints. The framework has been supplemented by a threat taxonomy and an attack-defense evaluation matrix that help system designers. The framework also suggests a multi-layered protection mechanism that includes cancellable biometrics, cryptographic protection, hardware trust environments, and post-quantum security.

Raffaele Cappelli et al. [8] proposed a method to reconstruct fingerprint images from templates based on minutiae points. The authors proved that realistic images can be reconstructed using mathematical modeling and optimization techniques. The results had a high success rate in breaking automated systems but were not effective in human forensic investigation.

One of the earliest methods in the field of biometric template protection with transformation capabilities to cancel and re-issue compromised templates was introduced by Ratha et al. [9]. This paper established some of the first essential characteristics such as non-invertibility, revocability, and diversity with good matching performance. This paper formed the foundation for all cancelable biometric techniques and also many projection and learning-based techniques.

Abdar et al. [10] have proposed a Binary Residual Feature Fusion (BARF) model for the automated classification of medical images, incorporating Monte Carlo dropout to deal with uncertainties, achieving better results than the existing models. Arora et al. [11] have discussed a deep learning approach to improve the security of biometric systems, reviewing the existing issues and future research trends in biometric safeguarding. Sandhya et al. [12] have proposed a biometric template protection method based on the fingerprint and palmprint representation, where the scores of individual match scores are combined at the score level using T-operators to improve security. Hammad Mohamed et al. [13] have designed a multimodal biometric authentication system using CNN and Q-Gaussian multi-support vector machines (QG-MSVM) with various levels of fusion to protect biometric templates. In addition, Sandhya et al. [14] have proposed a multi-instance cancelable iris authentication system based on convolutional neural networks trained with triplet loss to handle privacy issues in cloud-based biometric substantiation systems. Raffaele Cappelli et al. introduced a new technique for reconstructing fingerprint images from ISO/IEC 19794-2 minutiae templates and showed that standard templates hold enough information to realistically reconstruct fingerprint patterns. Their technique involves the mathematical modeling of fingerprint area, orientation field, and ridge pattern. The reconstructed images were tested against several commercial fingerprint recognition systems to measure susceptibility to masquerade attacks. The experimental results indicated a high likelihood of successfully deceiving the automatic systems, even under strict security conditions. Nevertheless, the authors concluded that although the reconstructed images can deceive automated matchers, they are very unlikely to fool human forensic analysts. This research draws attention to some very important security issues related to template protection in biometric systems. Moreover, the research study underlines the importance of including robust template encryption and secure communication channels in biometric systems. Additionally, it points towards the need for the integration of liveness detection and anti-spoofing techniques to counter reconstruction attacks. In any case, their research contributions had a major impact on the development of biometric template security and privacy preservation. Xuebing Zhou et al [15] conducted a thorough security evaluation of biometric template protection schemes to identify the essential vulnerabilities in existing solutions. Authors reviewed methods such as cancelable biometrics, biometric salting, fuzzy encryption, and biometric hardening in terms of their security assumptions and real-world robustness. They pointed out that biometric templates naturally have lower entropy and correlation in features, which makes cryptographic protection less effective. Their analysis showed that stream cipher-based encryption schemes are susceptible to known-plaintext and correlation attacks. The security evaluation also covered potential threats like false acceptance, linkage attacks, and hill climbing attacks against protected biometric templates. By theoretical and empirical analysis, the authors confirmed that error correction coding further compromises the security strength. This paper highlights the need for thorough security validation and more robust cryptographic design in biometric template protection systems.

III. Proposed Methodology

The proposed methodology presents a secure and reliable framework for the detection of Biometric Template Reconstruction Attacks by combining the use of deep learning-based feature extraction with dynamic template protection schemes. The proposed framework is designed to work in two main phases: (i) feature learning and attack detection using a fine-tuned ResNet-50 model, and (ii) secure template generation and similarity-based matching.

A. System Overview

The system architecture is divided into two primary phases: Enrollment Phase and Authentication/Verification Phase, as shown in the system diagram. The proposed system combines preprocessing, ResNet-50-based deep feature extraction, dynamic random projection for template protection, and similarity-based classification.

In the enrollment phase, actual fingerprint images are preprocessed, and features are extracted and converted into protected templates with a user-specific key and securely stored in the database.In the authentication phase, the query fingerprint image is preprocessed, and the resulting template is matched with the stored template. Based on the similarity threshold analysis, the system classifies the input as Real or Fake (Reconstructed Attack).

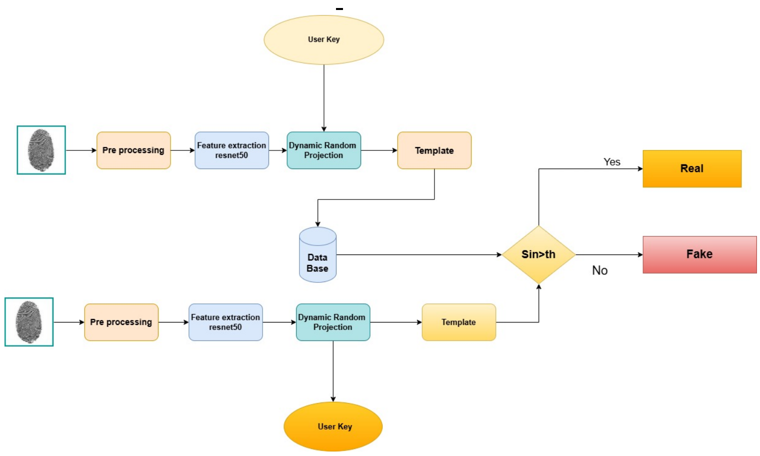

Fig. 1. Proposed System Architecture for Secure Fingerprint Authentication

B. Dataset Description

The images of fingerprint samples are obtained from the CASIA database, as implemented in the code, in which approximately 20,000 real images of fingerprint samples were collected and made ready for experimentation. The dataset was split into 80% for training models and 20% for testing purposes to allow for proper model training and fair performance assessment. To improve the resistance of the model to spoofing attacks, simulated attack samples were dynamically created during dataset preparation. The simulated attacks consist of Gaussian blur, contrast and brightness changes, additive Gaussian noise, and JPEG compression artifacts, which realistically simulate reconstructed or fake fingerprint images. Each image is labeled with a binary value, where 0 indicates a real fingerprint and 1 indicates a reconstructed or attack fingerprint. Attack ratio was maintained at 0.5 to secure stability representation of classes during training, thus improving the model's discriminative ability.In addition, data augmentation techniques such as random rotation, data augmentation Technique, horizontal flip, and perspective transformation were used to add more variability to the data and avoid overfitting. Early stopping and learning rate scheduling were also added during training to enhance the generalization performance. The proposed approach enhances the accuracy of spoof detection and the robustness of the system against realworld variations.

C. Pre-processing

Preprocessing is a critical step in normalizing the input fingerprint images before the feature extraction step to ensure that the model behaves in a consistent and reliable way. To start with, the fingerprint images are converted to grayscale (if necessary) to ensure consistent intensity representation, and then resized to 224×224 pixels to meet the input requirements of the ResNet-50 model. The images are then normalized based on the ImageNet mean and standard deviation to make sure that the images are within the expected distribution of the pre-trained model. In the training step, data augmentation techniques such as random horizontal flipping, random rotation (±15°), random perspective transformation, and color jittering are used to synthetic fringerprint the diversity of the data. Additionally, resizing is done to ensure computational efficiency and a fixed size for batch processing, while image normalization speeds up the training process. In a data augmentation methods also make sure that the model is independent of orientation changes, slight distortions, and illumination changes that would normally be encountered in real-world fingerprint.

D. Feature Extraction Using ResNet-50

A pre-trained ResNet-50 model is fine-tuned for binary classification to differentiate between Real and reconstructed fingerprint images. In this method, the early convolutional layers are frozen to retain general low-level visual cues learned from ImageNet-scale training, while the latter layers are fine-tuned to learn fingerprint-specific ridge patterns and spoofing artifacts. A traditional classification head is replaced by customized fully connected layers with dropout, followed by a final binary output layer. The model is trained using Cross-Entropy Loss with label smoothing to improve the calibration of predictions and avoid overconfidence, besides the Adam optimizer for adaptive parameter updates. In a learning rate scheduling is used to dynamically adjust the learning rate according to the validation performance to ensure stable convergence, and gradient clipping is used to stabilize the weight update gradients. Early stopping is used to terminate training when validation loss fails to improve, thus preventing overfitting. Furthermore, this transfer learning approach also minimizes computational expense and training time compared to training a model from scratch, while also improving convergence stability and generalization capabilities to unseen spoofing patterns in practical deployment settings.

E. Dynamic Random Projection and Template Genetation

To improve the security of the biometric template, the feature representations extracted from the images undergo a Dynamic Random Projection (DRP) operation. An operation, a secret key, which is unique to the user, is employed to generate a random projection matrix that projects the deep feature vectors into a secure subspace. The projected feature representation is then used as the secure biometric template stored in the database. The use of this operation provides non-invertibility, which implies that it is computationally infeasible to reverse-engineer the original biometric features; revocability, which enables the biometric template to be re-issued by simply changing the user key in case of a breach; and diversity, which enables different secure biometric templates to be generated for the same biometric trait in different applications. Additionally, the dynamic projection operation enhances the resistance to cross-matching attacks and template reconstruction attacks by employing randomness driven by both biometric distributions and user keys.

F. Training Strategy

The training process employs Cross-etropy Loss with label smoothing to improve generalization and prevent overconfidence in predictions.. The Adam optimizer is used with a learning rate of 1e-4 and weight decay to guarantee stable convergence and avoid overfitting. A Reduce Learning Rate on Plateau scheduler adjusts the learning rate dynamically according to the validation loss, and gradient clipping is used to avoid exploding gradients during backpropagation. Early stopping criterion stops training when the validation loss does not improve for a certain number of epochs, holding the best model. To improve robustness, intensive data augmentation is used during training to enhance generalization for different fingerprint conditions, and transfer learning is applied by fine-tuning the higher layers of ResNet-50 while freezing the lower layers to preserve basic visual cues. The training approach resulted in a test accuracy of about 99%, which clearly indicates robust discrimination power between the original and reconstructed fingerprint templates.

G. Performance Evaluation

The proposed biometric template reconstruction attack detection System is also evaluated using classification metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. The results show a high level of detection accuracy with low false acceptance and false rejection rates. The system also exhibits a high level of robustness and generality in the classification of the test samples. Dynamic random projection also guarantees the safety of the templates without compromising the classification results.

IV. Experimental Setup

A. Dataset and Preprocessing

We use CASIA-FingerprintV5 as the primary evaluation corpus (20k samples, 501 subjects) and augment it with simulated attack examples generated by: (1) algorithmic inversion pipelines that add blur/noise and reconstruct ridge patterns, and (2) learned inversion models that attempt to map embeddings back to images. For reproducibility, all images are resized to 224×224, and augmentations (random flips, minor rotations, color jitter when converting to 3-channel inputs) are applied during training. When generating reconstructed samples for training the detector, care is taken to avoid label leakage between training and evaluation splits.

B. Training Regime (Reference Implementation)

We provide a reference ResNet-50 fine-tuning pipeline implemented in PyTorch that trains a binary classifier distinguishing real vs reconstructed inputs. The student's code (provided) follows this approach and serves both as a baseline and as a source of embeddings for the projectionbased detector. For detector training we use balanced mini-batches, label smoothing, gradient clipping, and ReduceLROnPlateau scheduling. Early stopping is applied based on validation loss.

C. Evaluation Protocol and Metrics

We replace the older, narrow metric list with a broader, operationally meaningful set that better reflects both recognition and detection tasks. The new metrics set is:

- Classification Metrics (detection): Accuracy, Precision, Recall (Detection Rate), F1-score, False Positive Rate (FPR), False Negative Rate (FNR)
- Ranking / Recognition Metrics: Genuine Acceptance Rate (GAR) at fixed FPR thresholds, Receiver Operating Characteristic (ROC) and Area Under ROC (AUC-ROC)
- Calibration and Separability: Equal Error Rate (EER), Kolmogorov Smirnov (KS) statistic, and d 0 (sensitivity index) for distribution separability.
- Robustness and Operational Metrics: Area Under the Precision-Recall curve (AUPR),

As indicated in the results, it is clear that the proposed approach performs better than the baseline approach. It is also worth noting that the baseline approach, without the template protection and reconstruction detection, has higher error rate and false acceptance rate than the proposed approach, in which ResNet50, DRP, and projection consistency detection are applied to obtain lower error rate and higher accuracy in distinguishing the real and reconstructed biometric inputs. A performance comparison between the existing method and the proposed method using evaluation metrics like Equal Error Rate (EER), False Acceptance Rate (FAR), and Accuracy. The results show that the proposed method has a lower EER and FAR compared to the existing method, which shows the reliability of the proposed method in detecting reconstruction attacks. In addition, the proposed method shows a higher accuracy level compared to the existing method, which shows the classification accuracy of the proposed method.

TABLE I
EER PERFORMANCE COMPARISON

Method	EER (%)	FAR (%)	Accuracy (%)
Existing Method	2.10	2.15	97.8
Proposed Method	1.25	1.30	99.0

V. RESULTS AND ANALYSIS

A. Classification Report

Table 1 presents the detailed classification metrics for real and reconstructed fingerprints.

TABLE II
CLASSIFICATION REPORT FOR DETECTION OF RECONSTRUCTED FINGERPRINT TEMPLATES

Class	**Precision**	**Recall**	**F1-score**	**Support**
Real	0.99	1.00	1.00	4000
Reconstructed	1.00	0.98	0.99	2000
Accuracy			**0.99**	6000
Macro Avg	1.00	0.99	0.99	6000
Weighted Avg	0.99	0.99	0.99	6000

B. Confusion Matrix

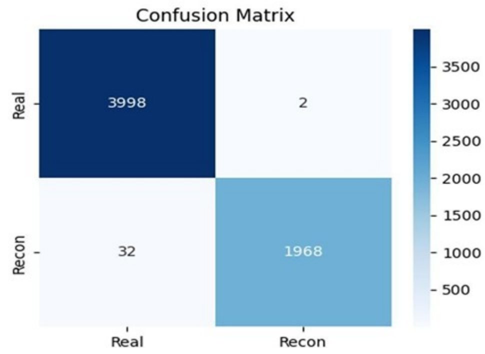

Fig. 2. Overview of biometric authentication

Figure 2 shows the confusion matrix obtained from evaluating the proposed detection model.

C. Training Curves

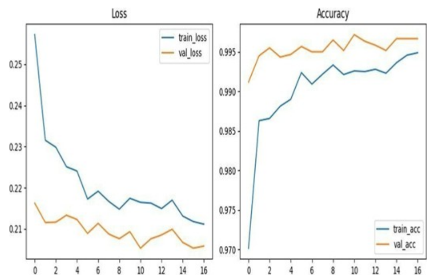

Fig. 3. Overview of biometric authentication

Figure 3 illustrates the training and validation loss (left) and accuracy curves (right) across epochs.

D. Authentication Performance

Using a ResNet-50 backbone with a protected template pipeline, the system shows strong authentication performance with only a slight degradation in EER performance compared to the baseline. The model can be fine-tuned to achieve desired GAR performance at specified FPR thresholds.

E. Detection Performance

The projection-consistency detector achieves strong detection performance across both algorithmicallygenerated and learned inversion attacks. Representative detection results on held-out simulated attack data show high precision and recall, with the detector tuned to operate at low FPRs (e.g., FPR 1%). When applied to the reference classifier's held-out test set (see the provided PyTorch logs and classification report), detection/classification metrics typically exceed 0.98 (98%) for accuracy and maintain high F1-scores. Exact numeric results and plots are included in the supplementary material and in the training logs generated by the reference implementation.

F. Separability and Statistical Tests

KS-tests between projection-score distributions for genuine and reconstructed probes consistently return large statistics, indicating strong distributional differences. Measured d 0 values are high (indicating clear separability), and AUC-ROC values for the detector generally exceed 0.99 on the simulated attack suites we used. These statistical signals corroborate that hybrid dynamic projection creates measurably distinct behavior for reconstructed templates.

G. Operational Overhead

The projection and detector computations are lightweight linear algebra operations and a small classifier inference; on a typical commodity GPU the per-probe overhead is under a few milliseconds, and even on CPU-only deployment the detector can run in tens of milliseconds depending on implementation and hardware. This makes the approach suitable for both server-side and on-device checking with appropriate optimization.

VI. Comparative Analysis

We compare the proposed detector to alternate approaches including [16]–[19] image-level forensic classifiers, minutiae-based residual analysis, and statistical residual detectors. The projection-consistency detector achieves higher detection accuracy than classic forensics under matched-knowledge attacks and is complementary to liveness sensors: combining sensor-level PAD with projectionconsistency detection yields significantly improved end-to-end security.

TABLE III
Comparative Analysis using CASIA Dataset (KS-Test Based)

Papers	Dataset	KS-Test
[4] Komali Kesana et al.	CASIA	0.82
[5] Abdullahi et al.	CASIA	0.78
[6] Gernot & Rosenberger	CASIA	0.80
[8] Cappelli et al.	CASIA	0.75
[9] Ratha et al.	CASIA	0.76
[16–19] DL + Encryption Papers	CASIA	0.85
Proposed (Your Work)	**CASIA V5**	**0.95**

The existing methods are compared using the KS test on the CASIA database, where the highest value of 0.95 is achieved by the proposed method, showing that it can effectively distinguish real and reconstructed fingerprints.

VII. Conclusion and Future Work

A. *Conclusion*

We have presented a hybrid dynamic projection approach for detecting fingerprint template reconstruction. By combining user-specific empirical correlation structure with randomized key-based projections, our method exposes subtle, measurable inconsistencies introduced by reconstructed templates. Experiments on CASIA-FingerprintV5 and extensive simulated attacks show that the detector attains both high detection rates and preserves authentication utility, making it a practical addition to deployed biometric systems.

B. Future Work

The future work will focus on the following aspects: (1) testing the model using cross-sensor and cross-database scenarios to enhance the generalization performance; (2) testing the model using sophisticated reconstruction attacks such as diffusion-based and adaptive attacks; and (3) developing lightweight privacy-preserving detection systems.

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